

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Create a platform that moves based upon desired amount of sunlight. The reason we have to create a platform that moves is because this platform will allow us to limit the amount of sunlight it gets and to move the shade or to put it back on.	We connected the motor to the roof then connected the battery to the motor which would move the roof.	I think we tried this about 20 times and it wasn't completely done there as it would be on and off working sometimes and sometimes not working.	Parts of the prototype may: — stop working — some parts might fall off	Fail: one or more parts of the prototype detaches Pass: zero parts of the prototype detach Goal: all parts of the prototype remain unchanged when the prototype is returned to its original position
2	Programmable for time of day and total time in the sun. The reason this is important is because it has to be able to handle the amount of sunlight that it gets and it has to be programmed to work for a certain amount of time. And this is really important because if it can't program for how long you want it to then it won't work to your capabilities.	We tested it at different times of the day. And different days with different temperatures on these days.	Throughout the plant shade project we tested this if I were to give an estimate probably around 30 times.	Parts of the prototype may: — stop working — The part isn't able to handle the heat	Fail: mean of the trials $\geq$ a part fails Pass: mean of the trials $<$ nothing goes wrong Goal: mean of the trials $<$ works for a long time and doesn't mess up

3	Accommodate a seedling size tray. The reason this is important is because it has to have enough space to hold the seedling and provide space because as the plant grows it holds more space.	Our base had to hold the plant and for this we found wood planks and connected some together to have a strong base and hold a lot of weight	Didn't really test this as we had a pretty good idea on how it would work and once we did it worked very well.	Parts of the prototype: —might break or fall of	Fail: part might fall off Pass: the base hold the plant very well and for a long time
4	Support up to 10lbs of weight	The base was made of wood and we connected it with nails and this made it hold a lot of weight	Didn't really test this as we had a pretty good idea on how it would work and once we did it worked very well.	The base might not hold the selected amount of weight	Fail: not holding the amount of weight Pass: The base holds the 10lb Goal: The base hold more than 10lb
5	Battery operated, works for 1 week without (solar) battery recharge	We put wires on it and connected it to the motor to make it work	Didn't really have trials as it didn't really fail	Works with acceptable amount of power	Fail: The motor burns out Pass: The motor works

